

QUESTIONS 19 / 19 completed

Plus

PLUS

Write a C++ program to overload the + operator to add two complex numbers.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Complex {
5 public:
6     int real, imag;
7
8     Complex() {
9         real = 0;
10        imag = 0;
11    }
12
13    Complex(int r, int i) {
14        real = r;
15        imag = i;
16    }
17
18    Complex operator + (Complex c2) {
19        Complex temp;
20        temp.real = real + c2.real;
21        temp.imag = imag + c2.imag;
```

OUTPUT

```
First complex number: 3 + 4i
Second complex number: 5 + 6i
Sum of complex numbers: 8 + 10i
```

INPUT

Your INPUT goes here!

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Increment

INCREMENT

Write a C++ program to overload the ++ operator to increment a variable.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Counter {
5 private:
6     int value;
7
8 public:
9     Counter(int v) : value(v) {}
10    Counter operator ++ () {
11        ++value;
12        return *this;
13    }
14    Counter operator ++ (int) {
15        Counter temp = *this;
16        value++;
17        return temp;
18    }
19
20    void display() {
21        cout << value << endl;
```

OUTPUT

```
6
```

INPUT

5

QUESTIONS 19 / 19 completed

<<

<<

Write a C++ program to overload the << operator to print the contents of a user-defined class.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Student {
6 private:
7     int id;
8     string name;
9     int marks;
10
11 public:
12     Student(int i, string n, int m) : id(i), name(n), marks(m) {}
13     friend ostream& operator<<(ostream& out, const Student& s) {
14         out << "ID: " << s.id << ", Name: " << s.name << ", Marks: " << s.marks;
15         return out;
16     }
17 };
18
19 int main() {
20     Student s1(1924, "Kuruva Ramesh", 95);
21     cout << s1;
```

OUTPUT

```
ID: 1924, Name: Kuruva Ramesh, Marks: 95
```

INPUT

QUESTIONS 19 / 19 completed

Equal

EQUAL

Write a C++ program to overload the == operator to compare two objects of a user-defined class.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Point {
5 private:
6     int x, y;
7
8 public:
9     Point(int x = 0, int y = 0) {
10         this->x = x;
11         this->y = y;
12     }
13
14     void setValues(int a, int b) {
15         x = a;
16         y = b;
17     }
18
19     bool operator==(const Point &p) {
20         return (x == p.x && y == p.y);
21     }
22 }
```

OUTPUT

Both points are equal

INPUT

1 2 1 2

QUESTIONS 19 / 19 completed

[]

[]

Write a C++ program to overload the [] operator to access the elements of an array using an index.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Array {
5 private:
6     int arr[10];
7 public:
8     void setValues() {
9         for (int i = 0; i < 10; i++) {
10             arr[i] = i * 10;
11         }
12     }
13
14     int& operator[](int index) {
15         return arr[index];
16     }
17 };
18
19 int main() {
20     Array obj;
21 }
```

OUTPUT

Enter index (0-9): Element at index 3 is: 30
After modification, element at index 3 is: 100

INPUT

3